

Business Insights Assessment

Medallion Data Pipeline · End-to-End AWS Architecture Reference

This document outlines the complete architectural design, execution steps, and metric layer definitions for the automated ELT data pipeline built using a **Medallion Architecture (Bronze → Silver → Gold)** on Amazon Web Services.

1. Technology Stack & Architecture

Layer / Component	AWS Service	Role & Function
Storage	Amazon S3 <code>business-insights-assessment-bucket</code>	Centralized repository for raw CSVs, cleaned Parquet files, Gold aggregates, and query results.
Catalog	AWS Glue Data Catalog	Tracks databases (<code>bia_db</code>) and external/managed table schemas for uniform querying.
Compute	Amazon Athena	Serverless Presto/Trino SQL engine executing all DDLs and transformations.
Orchestration	Apache Airflow (via <code>AthenaOperator</code>)	Automates execution sequences, DAG dependencies, scheduling, and error handling.

2. Medallion Data Flow

Data Flow: Raw CSVs in S3 → `bronze_dag` (External Tables) → `silver_dag` (Parquet Cleaned Tables) → `gold_dag` (7 Dedicated Business Metric Tables).

Bronze Layer (Raw Ingestion)

Registers raw CSV source files sitting in S3 as external SQL tables without altering source files. Includes type safety controls and null conversions for invalid records across 3 tables: `bronze_order_items`, `bronze_order_item_options`, and `bronze_date_dim`.

Silver Layer (Cleaned & Optimized)

Cleans strings, strips whitespace, filters out corrupt rows, safely casts timestamps using `TRY(CAST(...))`, and converts data into compressed, columnar **Parquet** format across 3 tables (`silver_order_items`, `silver_order_item_options`, `silver_date_dim`).

Gold Layer (7 Standalone Business Metric Tables)

Materialized aggregate tables designed for executive reporting and analytics:

- 1. `gold_customer_lifetime_value` : Customer monetary spend with N-Tile spending tiers.
- 2. `gold_customer_segmentation` : RFM scoring and behavior segmentation (VIP, New, Regular).
- 3. `gold_churn_indicators` : Recency metrics, order gaps, and "At-Risk" status flagging.
- 4. `gold_sales_trends` : Time-series aggregations by order date, restaurant, and category.
- 5. `gold_loyalty_program_impact` : Engagement analysis comparing loyalty members vs. non-members.
- 6. `gold_top_locations` : Store-level revenue performance and ranking.
- 7. `gold_pricing_discount_analysis` : Revenue impact of option discounts.

3. How to Run the Pipeline

```
# Step 1: Upload raw data to S3
aws s3 cp local_data/ s3://business-insights-assessment-bucket/raw/ --recursive

# Step 2: Deploy Airflow DAGs
aws s3 cp dags/bronze_dag.py s3://business-insights-assessment-bucket/dags/
aws s3 cp dags/silver_dag.py s3://business-insights-assessment-bucket/dags/
aws s3 cp dags/gold_dag.py s3://business-insights-assessment-bucket/dags/
```

Step 3: Open your Airflow UI, refresh DAGs, and trigger sequentially: `bronze_dag` → `silver_dag` → `gold_dag` .

Step 4: Verify in Athena using `SHOW TABLES IN bia_db;` and querying tables like `gold_top_locations` .

4. Cost Management & Cleanup

```
DROP DATABASE IF EXISTS bia_db CASCADE;
```

Empty and delete your S3 bucket, then terminate your MWAA environment to stop billing.